SDAIA
الهيئة السعودية للبيانات والذكاء الاصطناعي
Saudi Data & AI Authority

أكاديمية طويق
Tuwaiq Academy

DRAGO

Real-Time Detection Of Delivery Motorcycle Accidents Using AI

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PROBLEM STATMENT & SOLUSTIONS

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Problem statement:

By relying on motorcycles for delivery, traffic accidents on the roads increase, causing injuries and congestion.

Solution:

AI based model is proposed for real-time accident detection, which enhances road safety.

VISION 2030

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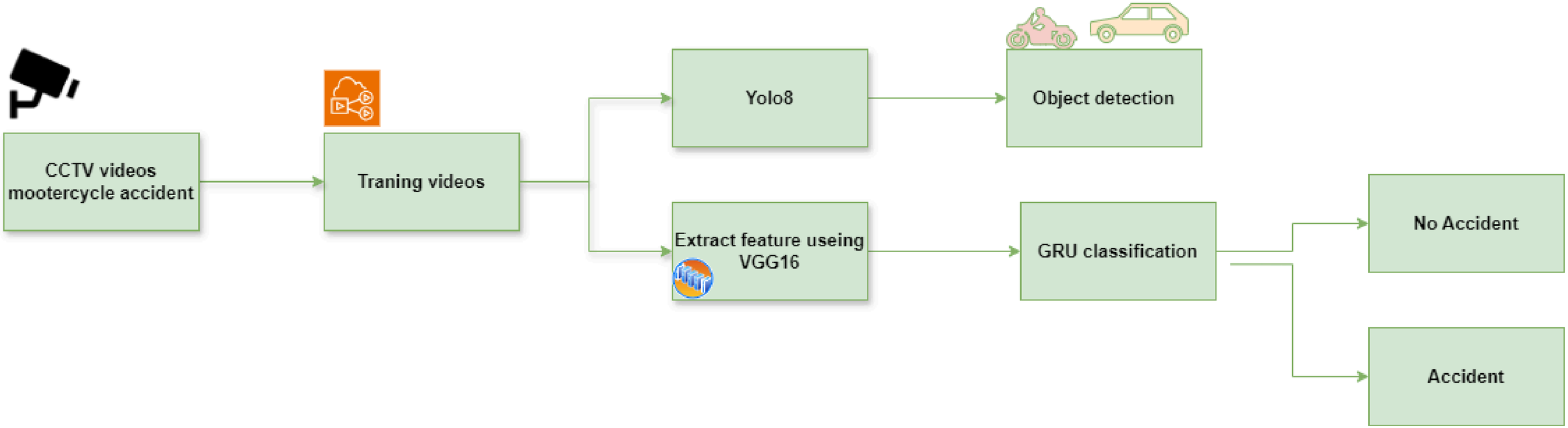
- **Urban Mobility & Public Transport: Reduces congestion and enhances transit efficiency. (Supports SDG 11)**
- **Economic Growth & Digital Transformation: Boosts logistics, business activities, and smart city initiatives. (Supports SDG 9)**
- **Environmental Sustainability & Safety: Lowers emissions and improves road safety. (Supports SDG 13 & SDG 3)**

Efficient traffic management improves urban mobility and safety, drives economic growth, and supports sustainability, contributing to a smarter, more resilient future aligned with Vision 2030 and the Sustainable Development Goals.

SOLUTION ARCHITECTURE



DATASETS

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Data video

30 videos from YouTube

Data images

RoboFlow dataset
Train set: 2642 images
Validation set: 195 images
Test set: 21 images

MODELS

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Object detection-YOLOV8

- High Detection Speed
- Comprehensive Object detection

Temporal Analysis for Accident Classification-GRU

- Handling Sequential Data
- Enhanced Classification Performance

Feature Extraction - VGG16

- extract visual features from detected motorcycles in video frames. These features will provide a detailed understanding of the motorcycle's condition

CHALLENGES

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Data Availability : Difficulty in sourcing relevant data for motorcycle accidents in Saudi Arabia.

Google Colab Limitations : Challenges with the Colab environment required upgrades.

Choose the appropriate model for sequential data : Determine which model gives the most accuracy classification

APPLICATIOMS

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Smart cities

- **Improved Road Safety**
- **Enhanced Traffic Management and Accident Prevention**

Cameras

- **Real-Time Monitoring and Detection**
- **Wide-Area Coverage and Accident Localization**

FUTURE WORK

Data expansion: Train the model on a larger dataset to improve accuracy and performance.

Large language model and Autonomous Agents: Write a report to analyze the incident by creating an agent that's write the report and send it by E-mail

PROJECT DEMO

Number of cars: 9

Number of motorcycle:2

Number of accideted detected: 1

Class value: 1